Diogo Ribeiro

ESMAD - Escola Superior de Media Arte e Design
Lead Data Scientist, Mysense.ai

dfr@esmad.ipp.pt 0009-0001-2022-7072

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Learning Goals

By the end of this session, you should be able to:

- connect deep learning, MLOps, explainability, and time series methods in one production workflow;
- choose methods based on business constraints (latency, risk, regulation, and reliability);
- identify common deployment and monitoring failure modes before they impact users.

Outline

1. Deep Learning Fundamentals
2. MLOps & Model Deployment
3. Explainable AI & Model Interpretability
4. Time Series Analysis & Forecasting

From Linear Models to Neural Networks

The evolution from simple to complex models

Linear Regression:

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + b$$

Limitations:

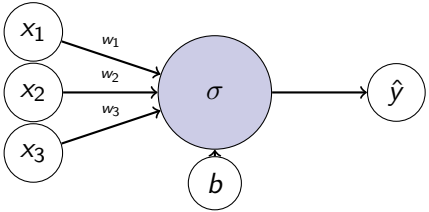
- Can only learn linear relationships
- Decision boundary is a hyperplane
- XOR problem unsolvable

Adding Non-Linearity:

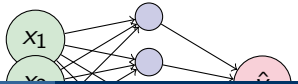
Key Insight: Compose linear transformations with non-linear activation functions!

$$\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + b)$$

Single Neuron (Perceptron):



Multi-Layer Perceptron (MLP):



Activation Functions

Non-linearities that make deep learning work

Why Activations Matter:

Without non-linearity, stacking layers is pointless:

$$f(W_2(W_1x)) = (W_2W_1)x = Wx$$

Still just linear!

Common Activations:

1. **Sigmoid:**

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Range: $(0, 1)$
Use: Output layer (binary classification)

Activation Function Comparison:

Function	Pros	Cons
Sigmoid	Smooth, interpretable	Vanishing gradient, not zero-cent
Tanh	Zero-centered	Vanishing gradient
ReLU	Fast, no vanishing gradient	Dead neurons (always 0)
Leaky ReLU	Fixes dead ReLU	Needs tuning
ELU	Smooth, negative values	Slower than ReLU
Swish	Smooth, self-gated	More computation

Modern Variants:

- **Leaky ReLU:**

$$f(z) = \begin{cases} z & z > 0 \\ 0.01z & z \leq 0 \end{cases}$$

Backpropagation: Training Neural Networks

The algorithm that makes deep learning possible

The Problem:
Given training data $\{(x^{(i)}, y^{(i)})\}_{i=1}^{n}$, find weights W that minimize loss:

$$\mathcal{L}(W) = \frac{1}{n}\sum_{i=1}^{n} \text{loss}(\hat{y}^{(i)}, y^{(i)})$$

The Solution: Gradient Descent

$$W \leftarrow W - \eta \nabla_W \mathcal{L}$$

But how to compute $\nabla_W \mathcal{L}$ for deep networks?

Backpropagation = Chain Rule
For a 2-layer network:

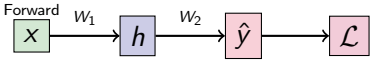
Forward Pass:

1. Compute activations layer by layer
2. Store intermediate values
3. Compute loss at output

Backward Pass:

1. Start from loss
2. Compute gradients backward
3. Update weights

Example: 1 Hidden Layer



Optimization Algorithms

Beyond vanilla gradient descent

Stochastic Gradient Descent (SGD):
Vanilla GD: Use all data

$$W \leftarrow W - \eta \nabla_W \mathcal{L}$$

SGD: Use mini-batches

$$W \leftarrow W - \eta \nabla_W \mathcal{L}_{\text{batch}}$$

Faster, more memory efficient, adds noise (helps escape local minima)

Modern Optimizers:

1. **SGD with Momentum:**

Optimizer Comparison:

Optimizer	Characteristics
SGD	Simple, requires good LR tuning
SGD+Momentum	Faster than SGD, less oscillation
RMSProp	Adaptive LR, good for RNNs
Adam	Adaptive LR, fast convergence, most popular
AdamW	Adam + weight decay (better regularization)
RAdam	Rectified Adam (warm-up built-in)

Learning Rate Schedules:

- **Step decay:** Reduce LR every N epochs
- **Exponential decay:** $\eta_t = \eta_0 e^{-kt}$
- **Cosine annealing:** Smooth reduction
- **1cycle:** Increase then decrease (fast.ai)

PyTorch Example:

Convolutional Neural Networks (CNNs)

Designed for spatial data (images, video)

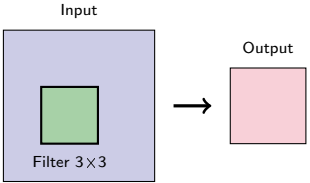
Why CNNs for Images?

- **Fully connected:**
  - 100×100 RGB image = 30K inputs
  - 1st hidden layer (1000 neurons) = 30M parameters!
- **Problems:**
  - Too many parameters
  - Ignores spatial structure
  - Not translation invariant

CNN Solutions:

1. **Local connectivity:**
   - Each neuron sees small region (receptive

Convolution Operation:



Filter slides across image:

$$(\text{img} * \text{filter})_{ij} = \sum_m \sum_n \text{img}_{i+m,j+n} \cdot \text{filter}_{mn}$$

Learned Filters Detect Features:

- Early layers: edges, corners

Modern CNN Architectures

Evolution from LeNet to EfficientNet
Classic Architectures:

1. **LeNet-5 (1998):**
   - 7 layers, digit recognition
   - First successful CNN
2. **AlexNet (2012):**
   - ImageNet winner
   - 8 layers, 60M parameters
   - ReLU, dropout, GPU training
3. **VGG (2014):**
   - 16-19 layers
   - Small 3×3 filters stacked
   - Very deep, simple architecture
4. **ResNet (2015):**
   - 50-152 layers
   - Skip connections!

Modern Developments:

Architecture	Innovation
Inception	Multiple filter sizes in parallel
ResNet	Skip connections (residual blocks)
DenseNet	Every layer connects to every other
MobileNet	Depthwise separable convs (mobile)
EfficientNet	Compound scaling (width+depth+resolution)
Vision Transformer	Self-attention instead of convolutions

Transfer Learning:

Instead of training from scratch:

1. Load pre-trained model (ImageNet)
2. Remove final layer

Recurrent Neural Networks (RNNs)

Networks with memory for sequential data

Why RNNs?

For sequences (text, time series, speech):

- Input length varies
- Temporal dependencies matter
- Order is important

RNN Idea:

Maintain hidden state h_t that captures history:

$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t)$$

$$y_t = W_{hy} h_t$$

Hidden state h_t is updated at each timestep.

LSTM (Long Short-Term Memory):

Solution to vanishing gradients:

- **Cell state:** Highway for information flow
- **Gates:** Control info flow
  - Forget gate: What to forget
  - Input gate: What to update
  - Output gate: What to output

LSTM Cell:

$$f_t = \sigma(W_f[h_{t-1}, x_t]) \quad \text{(forget)}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t]) \quad \text{(input)}$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t])$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

Transformers: Attention is All You Need

The architecture that revolutionized NLP and beyond

Limitations of RNNs/LSTMs:

- Sequential processing (slow)
- Still struggle with very long sequences
- Hard to parallelize

Transformer Innovation:

Replace recurrence with **self-attention**:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where:

- Q: Query (what I'm looking for)
- K: Key (what I have)

Transformer Architecture:

- **Encoder:** Process input sequence
- **Decoder:** Generate output sequence
- **Stack:** 6-12 layers typical

Famous Transformers:

Model	Task
BERT	Encoder-only, understanding
GPT	Decoder-only, generation
T5	Encoder-decoder, seq2seq
Vision Transformer (ViT)	Images as patches
CLIP	Vision-language alignment

Regularization in Deep Learning

Preventing overfitting in large models

The Overfitting Problem:

Deep networks have millions of parameters $\rightarrow$ can memorize training data!

Regularization Techniques:

1. **L2 Regularization (Weight Decay):**

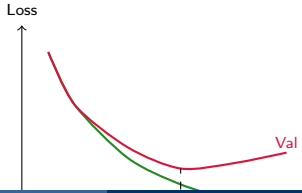
$$\mathcal{L}_{\text{total}} = \mathcal{L} + \lambda \sum_i w_i^2$$

   Penalizes large weights

2. **Dropout:**
   - Randomly set neurons to 0 (prob p)
   - Forces network to be robust
   - Ensemble effect
   - Typical: $p = 0.5$ for FC, $p = 0.2$ for

Early Stopping:

Monitor validation loss, stop when it starts increasing.



Knowledge Check: Deep Learning

Test your understanding

Question 1: Why don't we stack linear layers without activations?

A) It's too slow
B) **Multiple linear transformations collapse to one linear transformation**
C) It causes vanishing gradients
D) Backprop doesn't work

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A) Overfitting
B) **Vanishing gradients in very deep networks**
C) Slow training

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Summary & Best Practices

Key takeaways for deep learning

Architecture Choices:

- **Images:** CNNs (ResNet, EfficientNet) or Vision Transformers
- **Sequences:** Transformers (BERT, GPT) or LSTMs for small data
- **Tabular:** Often better with XGBoost/LightGBM
- **Time series:** LSTMs, Transformers, or specialized (N-BEATS)

Training Best Practices:

1. Start simple, add complexity

Common Pitfalls:

- × Not checking for bugs (use small dataset first)
- × Wrong loss function
- × Learning rate too high/low
- × Not shuffling training data
- × Using test set for validation
- × Forgetting to normalize inputs
- × No baseline comparison

Resources:

- **Courses:**
  - Andrew Ng: Deep Learning Specialization

The ML Production Gap

Why 87% of ML projects never make it to production

Research vs. Production:

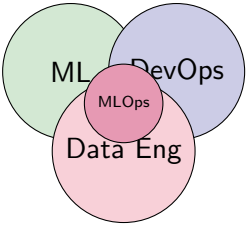
Research/Notebook	Production
Static dataset	Live, changing data
Batch processing	Real-time inference
Single model version	Multiple versions
Local machine	Distributed systems
Manual execution	Automated pipelines
No monitoring	24/7 monitoring
Reproduce once	Reproduce always

Challenges in Production:

1. Data drift: Input distribution changes
2. Concept drift: Relationship changes

What is MLOps?

ML + DevOps + Data Engineering



MLOps Goals:

- **Automation**: Reduce manual work
- **Reproducibility**: Same inputs → same outputs

The ML Lifecycle

From data to deployment and back

Traditional Software:

$$\text{Code} \rightarrow \text{Build} \rightarrow \text{Deploy}$$

Machine Learning:

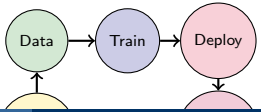
$$\text{Code} + \text{Data} + \text{Config} \rightarrow \text{Model} \rightarrow \text{Deploy}$$

More complex! Models are artifacts built from code AND data.

Complete ML Lifecycle:

1. **Data Collection**
   - Gather raw data
   - Set up pipelines
5. **Model Deployment**
   - Serve predictions
   - A/B testing
6. **Monitoring**
   - Track performance
   - Detect drift
7. **Retraining**
   - Update with new data
   - Close the loop

It's a Cycle, Not Linear!



Experiment Tracking

Managing the chaos of ML experiments

The Problem:

- Tried 50 experiments
- Which hyperparameters worked best?
- Can't reproduce last week's results
- Lost track of what changed

What to Track:

1. **Code:**
   - Git commit hash

MLflow Example:

```
import mlflow

mlflow.set_experiment("churn_prediction

with mlflow.start_run():
    # Log parameters
    mlflow.log_param("n_estimators", 10
    mlflow.log_param("max_depth", 10)

    # Train model
    model = RandomForestClassifier(
        n_estimators=100,
        max_depth=10
```

Model and Data Versioning

Git for machine learning assets

Why Version Everything?

- **Reproducibility**: Recreate any experiment
- **Debugging**: What changed between v1 and v2?
- **Rollback**: Revert to previous version
- **Compliance**: Audit trail for regulations
- **Collaboration**: Team knows what's happening

Data Version Control (DVC):

DVC Benefits:

- Data stored in cloud (S3, GCS, Azure)
- Git tracks only metadata
- Lightweight repositories
- Easy collaboration
- Pipeline versioning

ML Pipeline with DVC:

```
# dvc.yaml
stages:
  preprocess:
    cmd: python preprocess.py
    deps:
      data/raw
```

Deployment Patterns

How to serve model predictions

1. **Batch Prediction:**
   - Process data in batches (hourly, daily)
   - Generate predictions offline
   - Store in database
   - Serve pre-computed predictions

Use when:

✓ Don't need real-time

✓ Can predict in advance

✓ Large-scale scoring

Flask API Example:

```
from flask import Flask, request, jsoni
import joblib

app = Flask(__name__)
model = joblib.load('model.pkl')

@app.route('/predict', methods=['POST']
def predict():
    data = request.get_json()
    features = data['features']

    prediction = model.predict([feature
    probability = model.predict_proba([
```

Containerization with Docker

Package model + dependencies + code together

Why Docker?

- **"Works on my machine"** → Works everywhere!
- Reproducible environments
- Easy deployment
- Isolated dependencies
- Scalable (Kubernetes)

Dockerfile Example:

```
FROM python:3.9-slim

# Set working directory
WORKDIR /app
```

Build and Run:

```
# Build image
docker build -t my-ml-model:v1 .

# Run container
docker run -p 5000:5000 my-ml-model:v1

# Test
curl -X POST http://localhost:5000/pred
  -H "Content-Type: application/json" \
  -d '{"features": [1.0, 2.0, 3.0]}'
```

Docker Compose (Multi-Container):

```
# docker-compose.yml
```

Model Monitoring in Production

What to monitor and why

Why Monitor?

Models degrade over time!

- Data distribution shifts
- Concept drift (relationships change)
- Bugs in production pipeline
- Adversarial attacks

What to Monitor:

1. **Data Quality**
   - Missing values
   - Out-of-range values
   - Schema changes
2. **Data Drift**
5. **System Metrics**
   - Latency (p50, p95, p99)
   - Throughput (requests/sec)
   - Error rate
   - Resource usage (CPU, memory)

Example: Data Drift Detection

```
from evidently.dashboard import Dashboa
from evidently.tabs import DataDriftTab

dashboard = Dashboard(
    tabs=[DataDriftTab()]
)
```

CI/CD for Machine Learning

Continuous Integration and Deployment

Traditional CI/CD:

1. Code commit → Git
2. Run tests
3. Build artifact
4. Deploy to production

ML CI/CD Additions:

- **CT:** Continuous Training
- **CD:** Continuous Deployment
- **CM:** Continuous Monitoring

GitHub Actions Example:

```
# .github/workflows/ml-pipeline.yml
deploy:
  needs: train
  if: github.ref == 'refs/heads/main'
  runs-on: ubuntu-latest
  steps:
    - name: Build Docker image
      run: docker build -t model:${{
          github.sha }} .

    - name: Push to registry
```

Knowledge Check: MLOps

Test your understanding

Question 1: Your model was 92% accurate in development but 78% in production. What's likely the issue?

A) The model is overfitted
B) Wrong metrics used
C) **Data drift - production data differs from training data**
D) Need more training data

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Explanation

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A) Batch prediction (daily)

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Summary & Best Practices

Building production ML systems
The MLOps Checklist:

1. **Version everything**
   - Code (Git)
   - Data (DVC)
   - Models (MLflow)
   - Environment (Docker)
2. **Automate pipelines**
   - Data preprocessing
   - Model training
   - Evaluation
   - Deployment
3. **Test rigorously**
   - Unit tests
   - Integration tests
   - Data validation
   - Model performance

Common Pitfalls:

- × Not monitoring after deployment
- × No rollback plan
- × Training-serving skew
- × No data validation
- × Ignoring latency requirements
- × Not documenting decisions

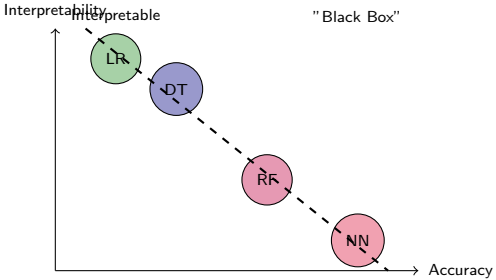
Tools Ecosystem:

Category	Tools
Experiment Tracking	MLflow, W&B
Versioning	DVC, Git LFS
Model Serving	TensorFlow Serving, TorchServe
Containerization	Docker, Kubernetes

Why Interpretability Matters

The black box problem in machine learning

The Accuracy-Interpretability Tradeoff:



Traditional View:

- Simple models = interpretable but less accurate
- Complex models = accurate but black box

Why We Need Interpretability:

1. **Trust & Adoption**
   - Doctors won't use unexplainable diagnosis
   - Banks must explain loan decisions
2. **Debugging & Improvement**
   - Identify when model fails
   - Find data leakage
   - Improve features
3. **Regulatory Compliance**
   - GDPR "right to explanation"
   - Fair lending laws
   - Medical device approval
4. **Fairness & Ethics**
   - Detect bias

Types of Interpretability

Taxonomy of explanation methods

Intrinsic vs. Post-hoc:

Intrinsic	Post-hoc
Model is inherently interpretable	Apply to any model after training
Linear regression, decision trees	SHAP, LIME
Limited complexity	Can explain black boxes

Model-Specific vs. Model-Agnostic:

Model-Specific	Model-Agnostic
Uses model internals	Treats model as black box
Attention weights, neuron activations	LIME, PDP
More accurate	Works for any model

Global vs. Local:

- **Global:** How model works overall
  - Feature importance

Levels of Interpretability:

1. **Algorithm Transparency**
   - How does the algorithm work?
   - Mathematical properties
2. **Global Model Interpretability**
   - What did the model learn overall?
   - Feature importances
   - Decision rules
3. **Local Interpretability**
   - Why this specific prediction?
   - Which features mattered for this instance?

The Interpretability Toolbox:

Method	Scope	Type

Intrinsically Interpretable Models

Models we can directly understand

Linear Regression:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Interpretation:

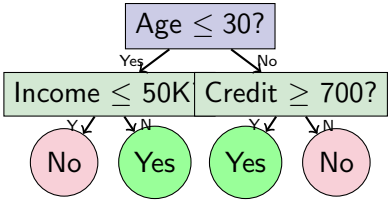
- β_j: Change in y for 1-unit change in x_j
- Sign: positive/negative relationship
- Magnitude: strength of effect

Caveats:

- Assumes linearity
- Correlation $\neq$ causation
- Multicollinearity issues

Logistic Regression:

Decision Trees:



Interpretation:

- Clear if-then rules
- Easy to visualize
- Mimics human decision-making

Limitations:

- High variance (unstable)

SHAP: Unified Framework for Interpretability

Game theory meets machine learning

The Core Idea:

How much does each feature contribute to prediction?

Based on Shapley Values from Game Theory:

Consider prediction as "coalition game":

- Players = features
- Payoff = prediction
- Fair distribution of credit

SHAP Value for Feature i:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

SHAP in Practice:

```
import shap

# Train model
model = XGBClassifier()
model.fit(X_train, y_train)

# Create explainer
explainer = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_t

# Visualize
shap.summary_plot(shap_values, X_test)
shap.force_plot(explainer.expected_valu
```

LIME: Local Interpretable Model-Agnostic Explanations

Explain any model locally with simple approximations

The LIME Approach:

Key Idea: Approximate complex model locally with interpretable model

Algorithm:

1. Select instance to explain
2. Generate perturbed samples nearby
3. Get predictions from black box
4. Train simple model (linear/tree) on perturbed data
5. Use simple model to explain

LIME Implementation:

```
from lime import lime_tabular

# Create explainer
explainer = lime_tabular.
    LimeTabularExplainer(
        X_train,
        feature_names=feature_names,
        class_names=['No', 'Yes'],
        mode='classification'
    )

# Explain instance
explanation = explainer.explain_instanc
```

Feature Importance Methods

Understanding what the model learned globally

1. Permutation Importance:

Shuffle feature j, measure performance drop:

```
from sklearn.inspection import
    permutation_importance

result = permutation_importance(
    model, X_test, y_test,
    n_repeats=10
)
```

Advantages:

- Model-agnostic
- Reliable
- Accounts for interactions

4. Partial Dependence Plots (PDP):

Show marginal effect of feature:

$$\hat{f}_S(x_S) = \mathbb{E}_{x_C}[\hat{f}(x_S, x_C)]$$

```
from sklearn.inspection import
    PartialDependenceDisplay

PartialDependenceDisplay.from_estimator
    model, X, features=[0, 1, (0,1)]
)
```

Shows:

- Feature effect on average

Counterfactual Explanations

"What would need to change for a different outcome?"

The Counterfactual Question:
"Why was my loan denied?"
Bad Answer: "Credit score was too low (SHAP value = -0.3)"
Better Answer (Counterfactual): "If your credit score were 680 instead of 620, you would be approved"

Properties of Good Counterfactuals:

1. **Proximity:** Close to original instance

Methods:

1. **Optimization-Based:**
   - Minimize distance subject to prediction constraint
   - Can use genetic algorithms
2. **DiCE** (Diverse Counterfactual Explanations):
   - Generate multiple diverse counterfactuals
   - Gives user options

Implementation:

```
import dice_ml

# Create DiCE explainer
dice = dice_ml.Dice(data, model)
```

Knowledge Check: Explainable AI

Test your understanding

Question 1: What is the key difference between SHAP and LIME?

A) SHAP is for trees only, LIME is model-agnostic
B) **SHAP has theoretical guarantees (Shapley values), LIME uses local approximation**
C) LIME is faster
D) They are the same

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Explanation

Answer: B - SHAP is based on game-theoretic Shapley values with mathematical guarantees (local accuracy, consistency). LIME approximates the model locally with a simpler model, which is fast but doesn't have the same theoretical properties. Both are model-agnostic, but SHAP provides more consistent explanations.

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Question 2: When would you use counterfactual explanations over feature importance?

A) When you need global model understanding
B) **When users need actionable advice on how to change outcomes**

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Summary & Best Practices

Building interpretable ML systems

The Interpretability Workflow:

1. **Model Selection:**
   - Start with interpretable baselines
   - Document accuracy-interpretability tradeoff
   - Justify complexity if needed
2. **Global Understanding:**
   - Permutation importance
   - Partial dependence plots
   - SHAP summary plots
3. **Local Explanations:**
   - SHAP force plots for key predictions

Common Mistakes:

- × Trusting explanations blindly
- × Using only one method
- × Ignoring feature correlations
- × Not validating with experts
- × Over-interpreting complex models

Tools & Resources:

- **Python Libraries:**
  - SHAP, LIME, ELI5
  - InterpretML (Microsoft)
  - AIX360 (IBM)
  - Captum (PyTorch)
- **Books:**

What is Time Series Data?

Data ordered by time with temporal dependencies

Definition:

A time series is a sequence of observations:

$$\{y_t : t = 1, 2, \ldots, T\}$$

where y_t is the value at time t.

Key Characteristic:

- Temporal ordering matters!
- Observations are *not* independent
- Past values influence future values
- Cannot shuffle observations

Examples:

- Stock prices (daily)

Time Series vs. Cross-Sectional:

Time Series	Cross-Sectional
Ordered by time	No time ordering
Temporal dependence	Independent obs
Cannot shuffle	Can shuffle
Train-test split: by time	Random split OK
Example: Daily stock price of AAPL	Example: House prices in a city

Types of Time Series:

1. **Univariate:** Single variable over time
   - Daily temperature
2. **Multivariate:** Multiple variables over time
   - Temperature, humidity, pressure

Time Series Components

Decomposing time series into interpretable parts

Additive Decomposition:

$$y_t = T_t + S_t + C_t + I_t$$

where:

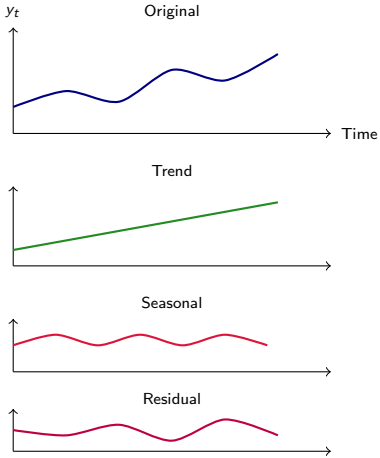
- T_t: **Trend** - Long-term direction
- S_t: **Seasonal** - Regular pattern
- C_t: **Cyclical** - Non-periodic fluctuations
- I_t: **Irregular** - Random noise

Multiplicative Decomposition:

$$y_t = T_t \times S_t \times C_t \times I_t$$

Use when seasonal variation increases with

Example: Airline Passengers



Stationarity: The Foundation

Why stationarity matters for time series modeling

Definition:
A time series is **strictly stationary** if:

$$P(y_{t_1}, \ldots, y_{t_k}) = P(y_{t_1+h}, \ldots, y_{t_k+h})$$

for all $t_1, \ldots, t_k$ and h.

Weak (Covariance) Stationarity:

1. $\mathbb{E}[y_t] = \mu$ (constant mean)
2. $\text{Var}[y_t] = \sigma^2$ (constant variance)
3. $\text{Cov}[y_t, y_{t-k}] = \gamma_k$ (depends only on lag k)

Testing for Stationarity:

Augmented Dickey-Fuller (ADF) Test

Tests null hypothesis of unit root:

$$H_0 : \text{Non-stationary (unit root)}$$

$$H_1 : \text{Stationary}$$

Reject H_0 if p-value < 0.05

Making Series Stationary:

1. **Differencing:**

$$\Delta y_t = y_t - y_{t-1}$$

ARIMA Models

AutoRegressive Integrated Moving Average
ARIMA(p, d, q) Components:

1. **AR(p):** AutoRegressive

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

   Uses past values

2. **I(d):** Integrated (Differencing)

$$\Delta^d y_t$$

   Makes stationary

3. **MA(q):** Moving Average

$$y_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}$$

   Uses past

Model Selection:

Tool	Purpose
ACF	Autocorrelation function - identifies MA(q)
PACF	Partial autocorrelation - identifies AR(p)
AIC/BIC	Information criteria - compare models
Auto-ARIMA	Automated search over (p,d,q) space

Seasonal ARIMA:

SARIMA(p,d,q)(P,D,Q)$_s$

Adds seasonal components with period s:

- (P, D, Q): Seasonal AR, I, MA
- s: Seasonal period (12 for monthly)

Python Implementation:

```
from statsmodels.tsa.arima.model import
ARIMA
```

Knowledge Check: Time Series Fundamentals

Multiple Choice Questions

Question 1: You have daily sales data for a retail store. Which decomposition would you likely use?

A) Additive, because sales don't vary
B) **Multiplicative, because seasonal variation grows with sales level**
C) Neither, sales data doesn't have seasonality
D) Use differencing instead

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Modern ML Methods for Time Series

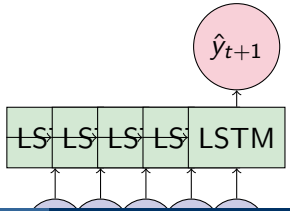
Beyond classical statistical approaches
Machine Learning Approaches:

1. **Feature Engineering + ML**
   - Lag features: $y_{t-1}, y_{t-2}, \ldots$
   - Rolling statistics: mean, std
   - Time features: day of week, month
   - Train XGBoost, Random Forest
2. **Prophet (Facebook)**
   - Additive model with trend + seasonality
   - Handles missing data, outliers
   - Easy to use, interpretable
3. **Neural Networks**
   - LSTM: Long Short-Term Memory
   - GRU: Gated Recurrent Unit
   - Transformer models
   - N-BEATS: Pure deep learning

When to Use What:

Method	Best For
ARIMA	Univariate, stationary, short-term
Exp Smoothing	Trends + seasonality, simple
Prophet	Business time series, holidays
XGBoost	Many features, non-linear
LSTM	Long sequences, complex patterns
Transformer	Multivariate, attention needed

Example: LSTM Architecture



Summary & Best Practices

Key takeaways for time series analysis

Core Principles:

1. **Temporal ordering matters**
   - Never shuffle time series data
   - Use time-based train/test splits
2. **Check stationarity**
   - ADF test before modeling
   - Difference if needed
3. **Understand your data**
   - Decompose into components
   - Identify seasonality, trend
   - Check for outliers, missing values
4. **Start simple, add complexity**

Common Pitfalls:

- × Using random train/test split
- × Ignoring non-stationarity
- × Data leakage from future
- × Over-differencing
- × Ignoring domain knowledge
- × Not handling seasonality
- × Using only RMSE (check MAE, MAPE too)

Resources:

- **Books:**
  - "Forecasting: Principles and Practice" (Hyndman & Athanasopoulos)

Integrated Takeaways

1. Model quality is necessary but insufficient: production success depends on deployment and monitoring discipline.
2. Explainability is part of product quality, not a post-launch add-on.
3. Forecasting and sequential decision systems require explicit handling of temporal structure and drift.
4. Strong teams combine statistical rigor, engineering practices, and domain context.

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Contact Information

Diogo Ribeiro

ESMAD - Escola Superior de Media Arte e Design
Lead Data Scientist, Mysense.ai

dfr@esmad.ipp.pt

0009-0001-2022-7072

github.com/diogoribeiro7

linkedin.com/in/diogoribeiro7

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